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S. No.	Topic	Page No.	Date	Grade/Marks	Signature
1.	learning all the basic R-programming		3/2/26		
2.	operating & decision making in R-programming		10/2/26		
3.	learn the basics of function in R & implement with examples.		12/2/26		
4.	Implement data frames in R write a program joins columns & rows in a data frame		12/2/26		
5.	Implementation different string manipulation in R		13/3/26		
6.	Implement different data Structures in R / vectors, lists, data frames.		23/3/26		
7.	program to read csv file and analyse data in csv file		12/3/26		
8.	Creating Piechart & Bar Charts using R		24/3/26		
9.	Creating data sets & statistical analysis on data		31/3/26		

//////////////////////////////////EXP-1//////////////////////////////////

```
> # Question 1: Student Information System
```

```
> roll_no <- 202
```

```
> name <- "Rahul"
```

```
> cgpa <- 9.1
```

```
> pass <- TRUE
```

```
> roll_no
```

```
[1] 202
```

```
> name
```

```
[1] "Rahul"
```

```
> cgpa
```

```
[1] 9.1
```

```
> pass
```

```
[1] TRUE
```

```
>
```

```
> # Question 2: Employee Salary Record
```

```
> emp_id <- 7002
```

```
> salary <- 55000.50
```

```
> department <- "HR"
```

```
> is_permanent <- FALSE
```

```
> emp_id
```

```
[1] 7002
```

```
> salary
```

```
[1] 55000.5
```

```
> department
```

```
[1] "HR"
```

```
> is_permanent
```

```
[1] FALSE
```

```
>
```

```
> # Question 3: Online Shopping Cart
```

```
> product_name <- "Mobile"
```

```
> price <- 25000
```

```
> stock_available <- 12
```

```
> discount_eligible <- TRUE
```

```
> product_name
```

```
[1] "Mobile"
```

```
> price
```

```
[1] 25000
```

```
> stock_available
```

```
[1] 12
```

```
> discount_eligible
```

```
[1] TRUE
```

```
> # Question 4: Weather Monitoring System
> city <- "Hyderabad"
> temperature <- 29.4
> rain <- TRUE
> wind_speed <- 8.6
> city
[1] "Hyderabad"
> temperature
[1] 29.4
> rain
[1] TRUE
> wind_speed
[1] 8.6
>
> # Question 5: Examination Result Processing
> subject <- "Physics"
> total_marks <- 100
> marks_obtained <- 91
> result <- marks_obtained >= 40
> subject
[1] "Physics"
> total_marks
[1] 100
> marks_obtained
[1] 91
> result
[1] TRUE
>
> # Question 6: Banking Application
> account_no <- 987654321
> holder_name <- "Priya"
> balance <- 80000.75
> account_active <- TRUE
> account_no
[1] 987654321
> holder_name
[1] "Priya"
> balance
[1] 80000.75
> account_active
[1] TRUE
```

```
> # Question 7: Movie Rating System
> movie_title <- "Avatar"
> rating <- 9.2
> reviews <- 4500
> available <- TRUE
> movie_title
[1] "Avatar"
> rating
[1] 9.2
> reviews
[1] 4500
> available
[1] TRUE
>
> # Question 8: Traffic Monitoring System
> vehicle_count <- 500
> location <- "City Center"
> avg_speed <- 35.4
> congested <- TRUE
> vehicle_count
[1] 500
> location
[1] "City Center"
> avg_speed
[1] 35.4
> congested
[1] TRUE
```


//////////////////////////////////EXP-2//////////////////////////////////

```
> # Question 1: Loan Eligibility Check
> age <- 30
> income <- 42000
> if(age > 21 & income > 30000){
+   print("Eligible for Loan")
+ } else {
+   print("Not Eligible for Loan")
+ }
[1] "Eligible for Loan"
>
> # Question 2: Attendance Requirement
> attendance <- 69
> if(attendance >= 75){
+   print("Eligible for Exam")
+ } else {
+   print("Not Eligible for Exam")
+ }
[1] "Not Eligible for Exam"
>
> # Question 3: Online Discount System
> amount <- 6500
> premium_member <- FALSE
> if(amount > 5000 | premium_member){
+   print("Discount Applied")
+ } else {
+   print("No Discount")
+ }
[1] "Discount Applied"
>
> # Question 4: Electricity Bill Slab
> units <- 350
> if(units < 100){
+   print("Low Usage")
+ } else if(units <= 300){
+   print("Medium Usage")
+ } else {
+   print("High Usage")
+ }
[1] "High Usage"
```



```
> # Question 5: Employee Bonus Eligibility
> experience <- 8
> rating <- 4.8
> if(experience > 5 & rating > 4){
+   print("Bonus Eligible")
+ } else {
+   print("Not Eligible for Bonus")
+ }
[1] "Bonus Eligible"
>
> # Question 6: Password Validation
> password <- "mypassword123"
> if(nchar(password) >= 8){
+   print("Valid Password")
+ } else {
+   print("Invalid Password")
+ }
[1] "Valid Password"
>
> # Question 7: Temperature Warning System
> temperature <- -3
> if(temperature < 0 | temperature > 40){
+   print("Extreme Temperature Warning")
+ } else {
+   print("Normal Temperature")
+ }
[1] "Extreme Temperature Warning"
>
> # Question 8: Grading System
> marks <- 95
> if(marks >= 90){
+   grade <- "A"
+ } else if(marks >= 75){
+   grade <- "B"
+ } else if(marks >= 50){
+   grade <- "C"
+ } else {
+   grade <- "Fail"
+ }
> print(grade)
[1] "A"
```



```
> # Question 9: Credit Card Approval System
> age <- 28
> income <- 50000
> credit_score <- 780
> if(age >= 21 & age <= 60 & income >= 25000 & credit_score >= 700){
+   print("Credit Card Approved")
+ } else {
+   print("Credit Card Rejected")
+ }
[1] "Credit Card Approved"
>
> # Question 10: Hospital Patient Risk Classification
> bp <- 130
> sugar <- 180
> if(bp > 140 & sugar > 200){
+   risk <- "High Risk"
+ } else if((bp >= 120 & bp <= 140) | (sugar >= 140 & sugar <= 200)){
+   risk <- "Medium Risk"
+ } else {
+   risk <- "Low Risk"
+ }
> print(risk)
[1] "Medium Risk"
>
> # Question 11: Smart Traffic Signal Control
> vehicle_count <- 250
> if(vehicle_count > 200){
+   print("Long Green Signal")
+ } else if(vehicle_count >= 100){
+   print("Medium Green Signal")
+ } else {
+   print("Short Green Signal")
+ }
[1] "Long Green Signal"
```



```
# Question 12: Employee Performance Appraisal System
attendance <- 94
performance <- 9
project_completed <- TRUE
if(attendance >= 90 & performance >= 8 & project_completed){
  print("Eligible for Appraisal")
} else {
  print("Not Eligible for Appraisal")
}
] "Eligible for Appraisal"
```

```
# Question 13: Online Exam Proctoring System
face_detected <- FALSE
multiple_faces <- FALSE
internet_connected <- TRUE
if(!face_detected | multiple_faces | !internet_connected){
  print("Exam Disqualified")
} else {
  print("Exam Allowed")
}
] "Exam Disqualified"
```

```
# Question 14: Dynamic Pricing in Ride-Hailing App
high_demand <- TRUE
available_drivers <- 20
if(high_demand & available_drivers < 50){
  print("Surge Pricing Applied")
} else {
  print("Normal Pricing")
}
] "Surge Pricing Applied"
```



```
. # Question 15: Industrial Machine Safety System
. temperature <- 85
. pressure <- 100
. vibration <- 5
. if(temperature > 80 | pressure > 120 | vibration > 7){
.   print("Machine Shutdown")
. } else {
.   print("Machine Running Normally")
. }
[1] "Machine Shutdown"
.
. # Question 16: University Admission Screening
. entrance_score <- 65
. twelfth_marks <- 80
. age <- 21
. if(entrance_score >= 70 & twelfth_marks >= 75 & age <= 25){
.   print("Admission Granted")
. } else {
.   print("Admission Rejected")
. }
[1] "Admission Rejected"
```


////////////////////////////////EXP-3////////////////////////////////

```
> # Question 1: Bank Interest Calculator
> simple_interest <- function(p, r, t){
-   si <- (p * r * t) / 100
-   return(si)
- }
> simple_interest(15000, 6, 3)
[1] 2700
>
> # Question 2: Student Result Processing
> check_result <- function(marks){
-   if(marks >= 40){
-       return("Pass")
-   } else {
-       return("Fail")
-   }
- }
> check_result(39)
[1] "Fail"
>
> # Question 3: Employee Salary Calculation
> net_salary <- function(basic_salary){
-   tax <- basic_salary * 0.10
-   net <- basic_salary - tax
-   return(net)
- }
> net_salary(45000)
[1] 40500
>
> # Question 4: Electricity Bill Generator
> electricity_bill <- function(units){
-   if(units <= 100){
-       bill <- units * 2
-   } else if(units <= 200){
-       bill <- (100 * 2) + (units - 100) * 3
-   } else {
-       bill <- (100 * 2) + (100 * 3) + (units - 200) * 5
-   }
-   return(bill)
- }
> electricity_bill(180)
[1] 440
```



```

> # Question 5: Online Shopping Discount
> final_amount <- function(amount){
+   if(amount >= 5000){
+     discount <- amount * 0.20
+   } else if(amount >= 2000){
+     discount <- amount * 0.10
+   } else {
+     discount <- 0
+   }
+   return(amount - discount)
+ }
> final_amount(5200)
[1] 4160
>
> # Question 6: Temperature Conversion System
> celsius_to_fahrenheit <- function(c){
+   f <- (c * 9/5) + 32
+   return(f)
+ }
> celsius_to_fahrenheit(25)
[1] 77
>
> # Question 7: Grading System
> assign_grade <- function(marks){
+   if(marks >= 90){
+     "A"
+   } else if(marks >= 75){
+     "B"
+   } else if(marks >= 50){
+     "C"
+   } else {
+     "Fail"
+   }
+ }
> assign_grade(49)
[1] "Fail"

```



```
> # Question 8: Voting Eligibility Checker
> check_voting <- function(age){
+   if(age >= 18){
+     return("Eligible to Vote")
+   } else {
+     return("Not Eligible to Vote")
+   }
+ }
> check_voting(21)
[1] "Eligible to Vote"
>
> # Question 9: Bank Loan EMI Calculator System
> calculate_emi <- function(principal, rate, years){
+   monthly_rate <- rate / (12 * 100)
+   months <- years * 12
+   emi <- (principal * monthly_rate * (1 + monthly_rate)^months) /
+     ((1 + monthly_rate)^months - 1)
+   return(round(emi, 2))
+ }
> calculate_emi(800000, 8.5, 15)
[1] 7877.92
>
> # Question 10: Hospital Patient Billing System
> hospital_bill <- function(consultation, room_charge, days){
+   subtotal <- consultation + (room_charge * days)
+   tax <- subtotal * 0.05
+   total <- subtotal + tax
+   return(total)
+ }
> hospital_bill(1000, 2500, 4)
[1] 11550
```



```
> # Question 11: Online Shopping Order Processing System
> order_total <- function(cart_value){
+   if(cart_value >= 3000){
+     discount <- cart_value * 0.10
+   } else {
+     discount <- 0
+   }
+   if(cart_value >= 5000){
+     delivery <- 0
+   } else {
+     delivery <- 100
+   }
+   final_amount <- cart_value - discount + delivery
+   return(final_amount)
+ }
> order_total(5500)
[1] 4950
>
> # Question 12: University GPA Calculation System
> calculate_gpa <- function(marks){
+   avg <- mean(marks)
+   if(avg >= 85){
+     result <- "Distinction"
+   } else if(avg >= 70){
+     result <- "First Class"
+   } else if(avg >= 50){
+     result <- "Second Class"
+   } else {
+     result <- "Fail"
+   }
+   return(result)
+ }
> calculate_gpa(c(90, 85, 88, 92, 87))
[1] "Distinction"
>
```



```
# Question 13: Smart Electricity Meter System
smart_meter_bill <- function(peak_units, offpeak_units){
  bill <- (peak_units * 6) + (offpeak_units * 3)
  total_units <- peak_units + offpeak_units
  if(total_units > 300){
    bill <- bill * 1.10
  }
  return(bill)
}
smart_meter_bill(200, 150)
1] 1815
```

```
# Question 14: Insurance Premium Calculation System
insurance_premium <- function(base, age, smoker){
  premium <- base
  if(age > 50){
    premium <- premium * 1.20
  }
  if(smoker){
    premium <- premium * 1.30
  }
  return(premium)
}
insurance_premium(6000, 55, TRUE)
1] 9360
```

```
# Question 15: Employee Payroll Processing System
payroll <- function(basic){
  hra <- basic * 0.20
  da <- basic * 0.10
  gross <- basic + hra + da
  tax <- gross * 0.12
  net <- gross - tax
  return(net)
}
payroll(50000)
.] 57200
```

```
# Question 16: Smart City Water Usage Monitoring
water_usage_status <- function(litres){
  if(litres <= 500){
    "Normal"
  } else if(litres <= 1000){
    "High Usage"
  } else {
    "Critical"
  }
}
water_usage_status(1200)
.] "Critical"
```


////////////////////////////////////EXP-4////////////////////////////////////

Question 1: College Student Record System

```
students <- data.frame(  
  RollNo = c(201, 202, 203),  
  Name = c("Rahul", "Priya", "Kiran"),  
  Dept = c("CSE", "ECE", "IT")  
)
```

```
marks <- data.frame(  
  Marks = c(88, 91, 79)  
)
```

```
student_data <- cbind(students, marks)  
student_data
```

RollNo	Name	Dept	Marks
201	Rahul	CSE	88
202	Priya	ECE	91
203	Kiran	IT	79

```
new_students <- data.frame(  
  RollNo = c(204),  
  Name = c("Anjali"),  
  Dept = c("CSE"),  
  Marks = c(85)  
)
```

```
final_data <- rbind(student_data, new_students)  
final_data
```

RollNo	Name	Dept	Marks
201	Rahul	CSE	88
202	Priya	ECE	91
203	Kiran	IT	79
204	Anjali	CSE	85


```

# Question 2: Employee Management System
employee_details <- data.frame(
  EmpID = c(11, 12, 13),
  Name = c("Aman", "Sneha", "Ravi"),
  Dept = c("HR", "IT", "Finance")
)

salary_details <- data.frame(
  Salary = c(40000, 55000, 60000)
)

employees <- cbind(employee_details, salary_details)
employees

```

EmpID	Name	Dept	Salary
11	Aman	HR	40000
12	Sneha	IT	55000
13	Ravi	Finance	60000

```

new_employee <- data.frame(
  EmpID = 14,
  Name = "Neha",
  Dept = "IT",
  Salary = 58000
)

employees <- rbind(employees, new_employee)
employees

```

EmpID	Name	Dept	Salary
11	Aman	HR	40000
12	Sneha	IT	55000
13	Ravi	Finance	60000
14	Neha	IT	58000


```

# Question 3: Hospital Patient Records
patient_info <- data.frame(
  PatientID = c(301, 302),
  Name = c("Suresh", "Lakshmi"),
  Age = c(50, 40)
)

treatment_cost <- data.frame(
  Cost = c(18000, 22000)
)

patient_records <- cbind(patient_info, treatment_cost)
patient_records

```

PatientID	Name	Age	Cost
301	Suresh	50	18000
302	Lakshmi	40	22000

```

new_patient <- data.frame(
  PatientID = 303,
  Name = "Arjun",
  Age = 45,
  Cost = 20000
)

patient_records <- rbind(patient_records, new_patient)
patient_records

```

PatientID	Name	Age	Cost
301	Suresh	50	18000
302	Lakshmi	40	22000
303	Arjun	45	20000


```
# Question 4: Product Inventory System
products <- data.frame(
  ProductID = c(1, 2, 3),
  ProductName = c("Pen", "Book", "Bottle")
)
```

```
stock <- data.frame(
  Quantity = c(100, 60, 40)
)
```

```
inventory <- cbind(products, stock)
inventory
```

ProductID	ProductName	Quantity
1	Pen	100
2	Book	60
3	Bottle	40

```
new_product <- data.frame(
  ProductID = 4,
  ProductName = "Bag",
  Quantity = 30
)
```

```
inventory <- rbind(inventory, new_product)
inventory
```

ProductID	ProductName	Quantity
1	Pen	100
2	Book	60
3	Bottle	40
4	Bag	30

|

////////////////////////////////EXP-5////////////////////////////////

```
# Question 1: Employee Name Standardization
names <- c("arun", "meena", "vijay")
upper_names <- toupper(names)
name_length <- nchar(upper_names)
upper_names
1] "ARUN" "MEENA" "VIJAY"
name_length
1] 4 5 5

# Question 2: Student Email Processing
emails <- c("arun@gmail.com", "meena@yahoo.com", "vijay@outlook.com")
usernames <- sub("@.*", "", emails)
usernames
1] "arun" "meena" "vijay"

# Question 3: Online Form Input Cleanup
input <- c(" Python ", " Machine Learning ", " Cloud ")
clean_input <- trimws(input)
clean_input
1] "Python" "Machine Learning" "Cloud"

# Question 4: Product Code Validation
product_codes <- c("PRD555", "PRD777", "XYZ999")
valid_codes <- startsWith(product_codes, "PRD")
valid_codes
1] TRUE TRUE FALSE

# Question 5: Text Replacement in Reports
report <- "error in login, error in payment"
updated_report <- gsub("error", "issue", report)
updated_report
1] "issue in login, issue in payment"

# Question 6: Password Strength Checker
password <- "welcome123"
if(nchar(password) >= 8){
  print("Strong Password")
} else {
  print("Weak Password")
}
1] "Strong Password"
```

```
# Question 7: Sentence Word Count
sentence <- "Python is easy to learn"
word_count <- length(strsplit(sentence, " ")[[1]])
word_count
1] 5
```

```
# Question 8: File Extension Extraction
files <- c("notes.txt", "video.mp4", "music.mp3")
extensions <- sub(".*\\. ", "", files)
extensions
1] "txt" "mp4" "mp3"
```

```
# Question 9: Student Name Abbreviation
names <- c("Suresh", "Lakshmi", "Deepika")
abbr <- substr(names, 1, 3)
abbr
1] "Sur" "Lak" "Dee"
```

```
# Question 10: Search Operation in Text
names <- c("Arun", "Bala", "Ravi", "John")
matched_names <- grep("a", names, value = TRUE)
matched_names
1] "Bala" "Ravi"
```


////////////////////////////////////EXP-6////////////////////////////////////

```
# Question 1: Student Marks Processing (Vectors)
```

```
marks <- c(65, 72, 88, 91, 79)
```

```
total <- sum(marks)
```

```
average <- mean(marks)
```

```
highest <- max(marks)
```

```
total
```

```
[1] 395
```

```
average
```

```
[1] 79
```

```
highest
```

```
[1] 91
```

```
# Question 2: Temperature Monitoring System (Vectors)
```

```
temperature <- c(28, 30, 29, 31, 32, 33, 30)
```

```
min_temp <- min(temperature)
```

```
max_temp <- max(temperature)
```

```
min_temp
```

```
[1] 28
```

```
max_temp
```

```
[1] 33
```

```
# Question 3: Employee Profile Storage (Lists)
```

```
employee <- list(
```

```
  ID = 205,
```

```
  Name = "Arjun",
```

```
  Salary = 55000,
```

```
  Department = "Finance"
```

```
)
```

```
employee
```

```
ID
```

```
[1] 205
```

```
Name
```

```
[1] "Arjun"
```

```
Salary
```

```
[1] 55000
```

```
Department
```

```
[1] "Finance"
```

```
employee$Name
```

```
[1] "Arjun"
```

```
employee$Salary
```

```
[1] 55000
```



```
# Question 4: Patient Medical Record System (Lists)
```

```
patient <- list(  
  PatientID = 305,  
  Name = "Meena",  
  Age = 50,  
  TestResults = c(110, 135, 128)  
)
```

```
patient
```

```
patientID
```

```
[1] 305
```

```
Name
```

```
[1] "Meena"
```

```
Age
```

```
[1] 50
```

```
TestResults
```

```
[1] 110 135 128
```

```
patient$TestResults
```

```
[1] 110 135 128
```

```
# Question 5: College Student Database (Data Frames)
```

```
students <- data.frame(  
  RollNo = c(11, 12, 13),  
  Name = c("Rahul", "Sneha", "Kiran"),  
  Dept = c("CSE", "ECE", "IT"),  
  Marks = c(78, 85, 92)  
)
```

```
students
```

```
RollNo Name Dept Marks  
11 Rahul CSE 78  
12 Sneha ECE 85  
13 Kiran IT 92
```

```
> # Question 6: Sales Report System (Data Frames)
```

```
> sales <- data.frame(  
+   Month = c("Apr", "May", "Jun"),  
+   Sales = c(45000, 52000, 61000)  
+ )
```

```
> total_sales <- sum(sales$Sales)
```

```
> total_sales
```

```
[1] 158000
```

```
> # Question 7: E-Commerce Order System
```

```
> prices <- c(999, 1499, 299)
```

```
> customer <- list(  
+   CustomerID = 401,  
+   Name = "Priya",  
+   City = "Hyderabad"  
+ )
```

```
> orders <- data.frame(  
+   Item = c("Mouse", "Keyboard", "USB"),  
+   Price = prices  
+ )
```

```
> prices
```

```
[1] 999 1499 299
```

```
> customer
```

```
CustomerID
```

```
[1] 401
```

```
Name
```

```
[1] "Priya"
```

```
City
```

```
[1] "Hyderabad"
```

```
> orders
```

```
Item Price  
Mouse 999  
Keyboard 1499  
USB 299
```



```
# Question 8: Online Exam System
```

```
subject_marks <- c(82, 76, 89)
```

```
student_profile <- list(
```

```
  Name = "Vijay",
```

```
  RollNo = 121,
```

```
  Marks = subject_marks
```

```
)
```

```
class_results <- data.frame(
```

```
  RollNo = c(111, 112, 113),
```

```
  Total = c(250, 240, 270)
```

```
)
```

```
student_profile
```

```
ame
```

```
] "Vijay"
```

```
ollNo
```

```
] 121
```

```
arks
```

```
] 82 76 89
```

```
class_results
```

```
RollNo Total
```

```
111 250
```

```
112 240
```

```
113 270
```

```
# Question 9: Bank Account Management
```

```
balances <- c(8000, 8500, 8200)
```

```
customer <- list(
```

```
  AccountNo = 56789,
```

```
  Name = "Lakshmi"
```

```
)
```

```
transactions <- data.frame(
```

```
  Date = c("10-01", "11-01", "12-01"),
```

```
  Amount = c(-1000, 2000, -500)
```

```
)
```

```
balances
```

```
1] 8000 8500 8200
```

```
customer
```

```
AccountNo
```

```
1] 56789
```

```
Name
```

```
1] "Lakshmi"
```

```
transactions
```

```
  Date Amount
```

```
10-01 -1000
```

```
11-01 2000
```

```
12-01 -500
```



```
> # Question 10: Real-World Summary Question
> scores <- c(78, 84, 91)
> profile <- list(
+   Name = "Suresh",
+   Age = 22,
+   Scores = scores
+ )
> records <- data.frame(
+   Subject = c("Math", "Science", "English"),
+   Marks = scores
+ )
> profile
$Name
[1] "Suresh"

$Age
[1] 22

$Scores
[1] 78 84 91

> records
  Subject Marks
1    Math    78
2 Science    84
3 English    91
```


////////////////////////////////EXP-7////////////////////////////////

```
# View first few rows
```

```
head(sales)
```

	Date	Product	Category	Quantity	Price
01-01-2026	Phone	Electronics	5	20000	
01-01-2026	Bag	Accessories	10	1500	
02-01-2026	Laptop	Electronics	3	50000	
02-01-2026	Headset	Electronics	7	2500	
03-01-2026	Bag	Accessories	5	1500	
03-01-2026	Phone	Electronics	3	20000	

```
# Check structure and summary
```

```
str(sales)
```

```
data.frame': 6 obs. of 5 variables:
```

```
$ Date : chr "01-01-2026" "01-01-2026" "02-01-2026" "02-01-2026" ...
```

```
$ Product : chr "Phone" "Bag" "Laptop" "Headset" ...
```

```
$ Category: chr "Electronics" "Accessories" "Electronics" "Electronics" ...
```

```
$ Quantity: int 5 10 3 7 5 3
```

```
$ Price : int 20000 1500 50000 2500 1500 20000
```

```
summary(sales)
```

Date	Product	Category	Quantity	Price
Length:6	Length:6	Length:6	Min. : 3.0	Min. : 1500
Class :character	Class :character	Class :character	1st Qu.: 3.5	1st Qu.: 1750
Mode :character	Mode :character	Mode :character	Median : 5.0	Median :11250
			Mean : 5.5	Mean :15917
			3rd Qu.: 6.5	3rd Qu.:20000
			Max. :10.0	Max. :50000

```
> # Total Sales per Product
```

```
> total_sales_product <- aggregate(TotalSales ~ Product, data=sales, sum)
```

```
> total_sales_product
```

	Product	TotalSales
1	Bag	22500
2	Headset	17500
3	Laptop	150000
4	Phone	160000

```
>
```

```
> # Product with Highest Sales
```

```
> top_product <- total_sales_product[which.max(total_sales_product$TotalSales), ]
```

```
> top_product
```

	Product	TotalSales
4	Phone	160000

```
>
```

```
> # Total Quantity Sold per Category
```

```
> quantity_per_category <- aggregate(Quantity ~ Category, data=sales, sum)
```

```
> quantity_per_category
```

	Category	Quantity
1	Accessories	15
2	Electronics	18

```
>
```

```
> # Filter High-Value Products
```

```
> high_sales <- total_sales_product[total_sales_product$TotalSales > 10000, ]
```

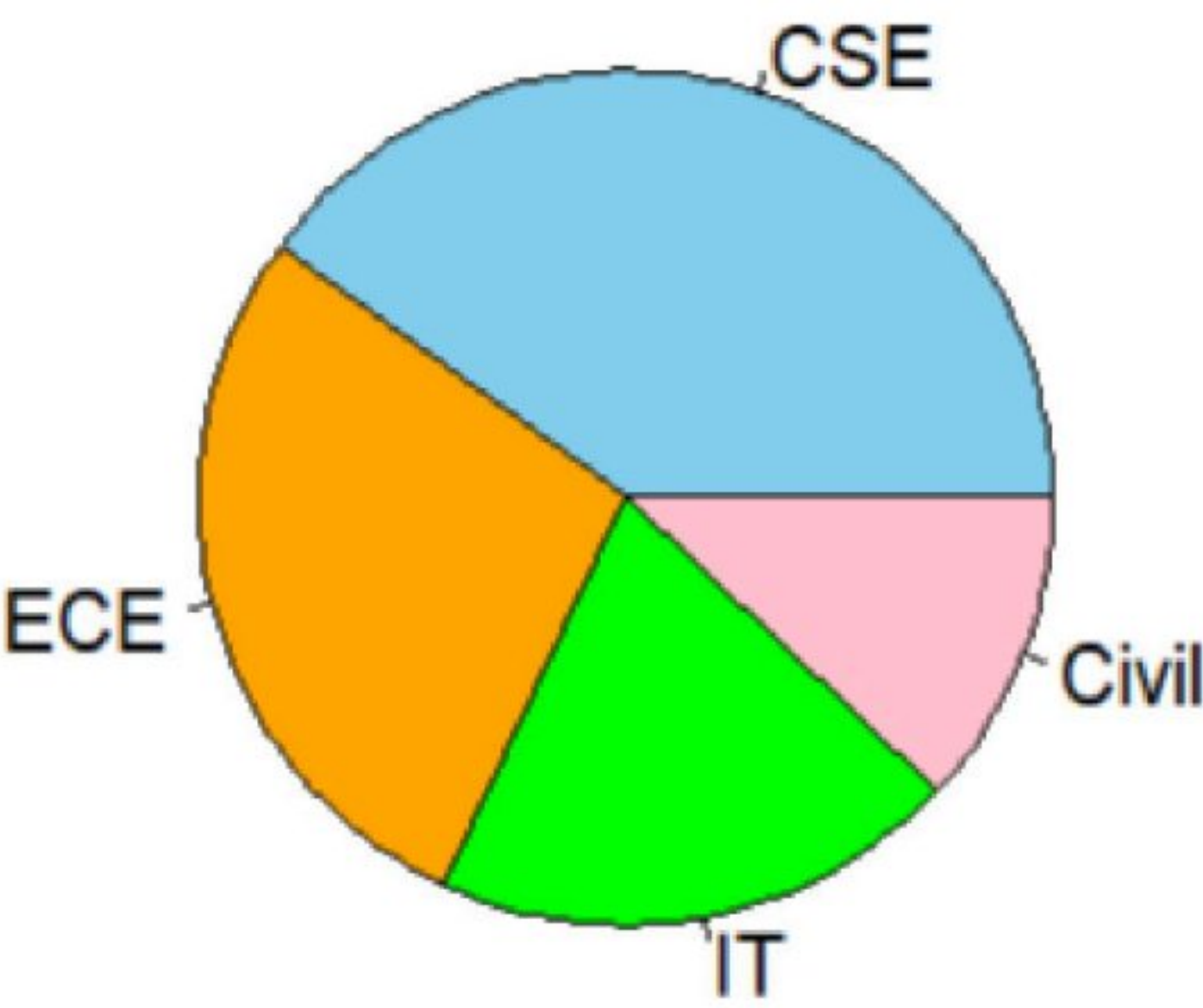
```
> high_sales
```

	Product	TotalSales
1	Bag	22500
2	Headset	17500
3	Laptop	150000
4	Phone	160000

Question 1: University Course Enrollment

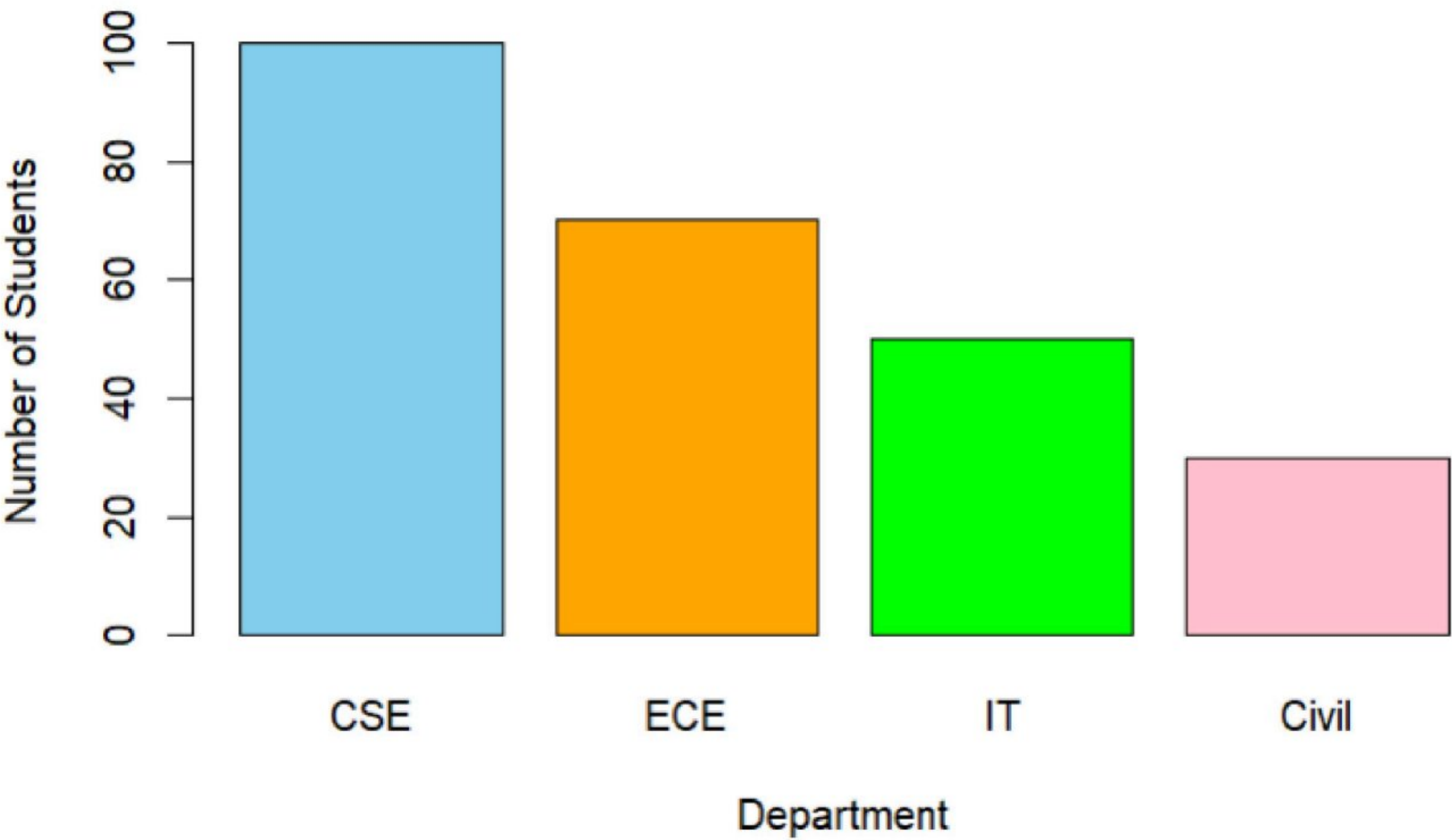
#PIECHART

Student Enrollment per Department



#BARCHART

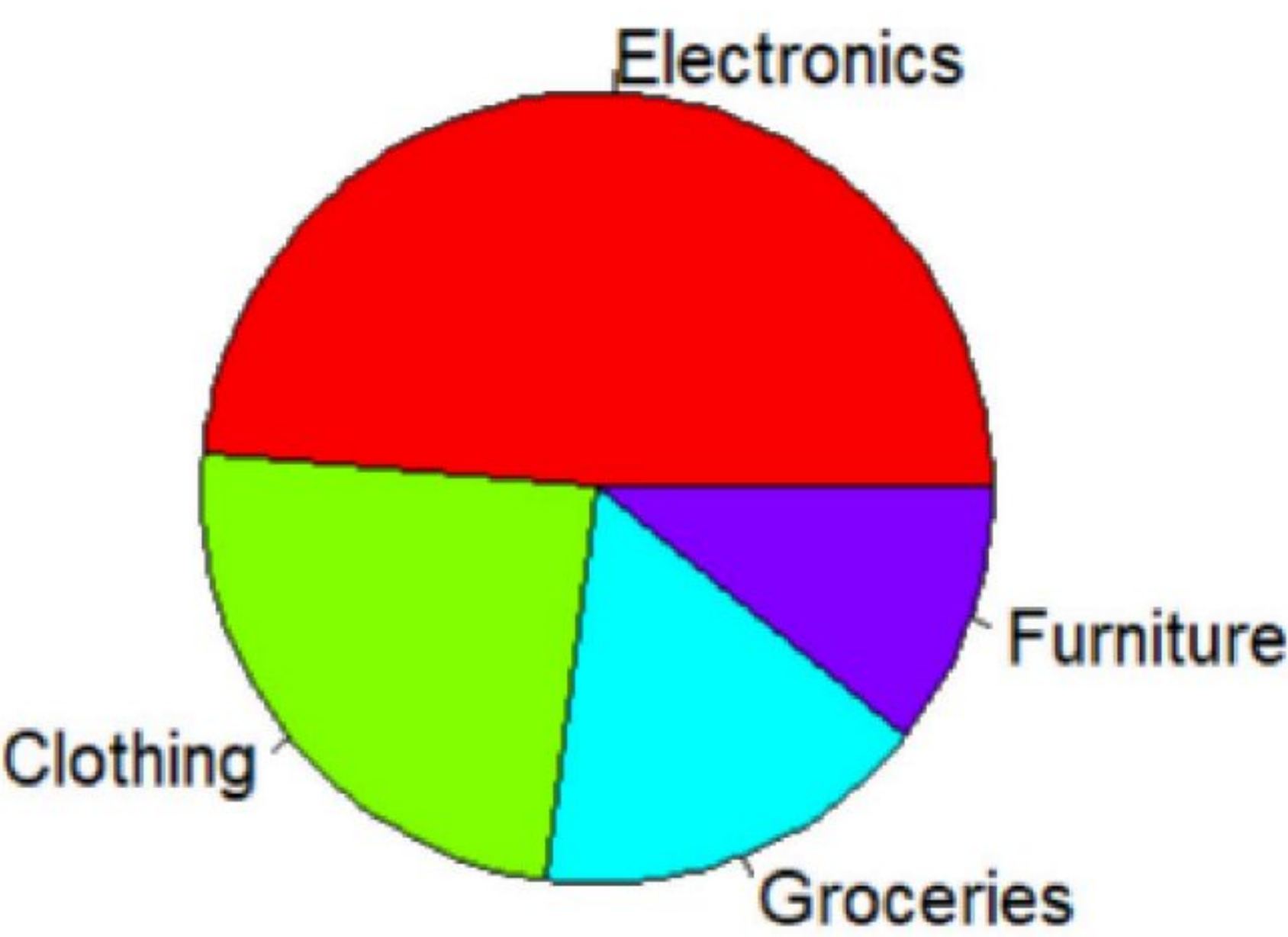
Number of Students per Department



Question 2: Retail Store Sales Distribution

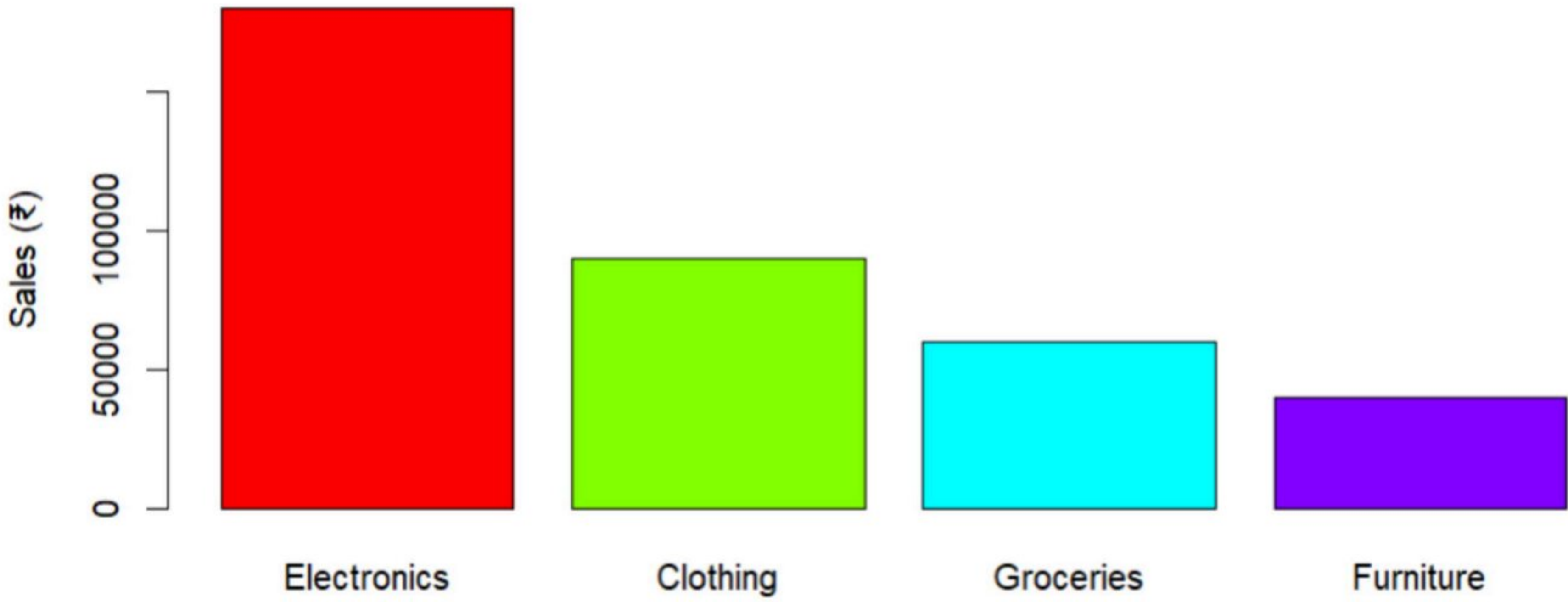
#PIECHART

Sales Distribution by Category



#BARCHART

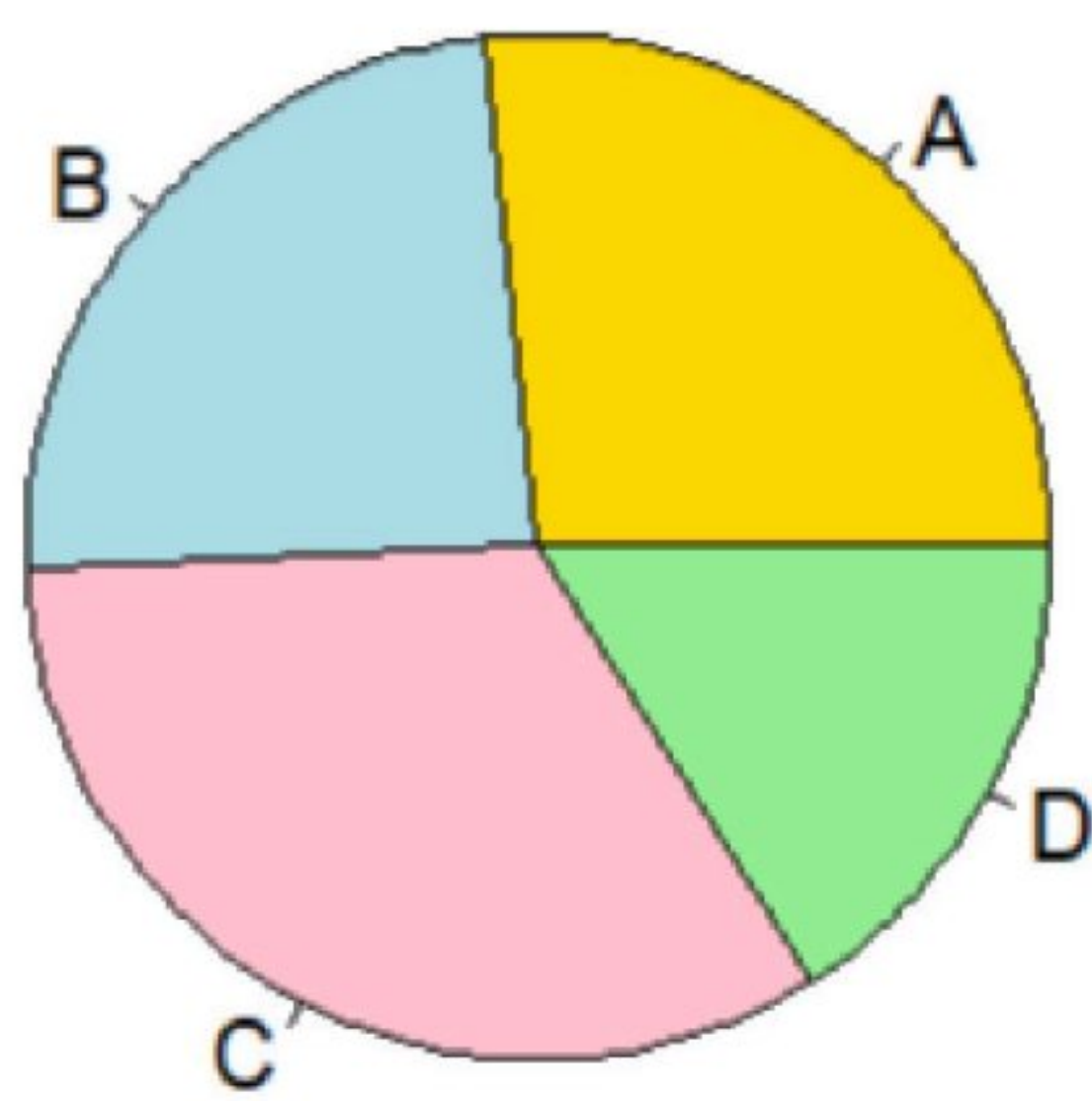
Sales per Category



Question 3: Online Exam Result Analysis

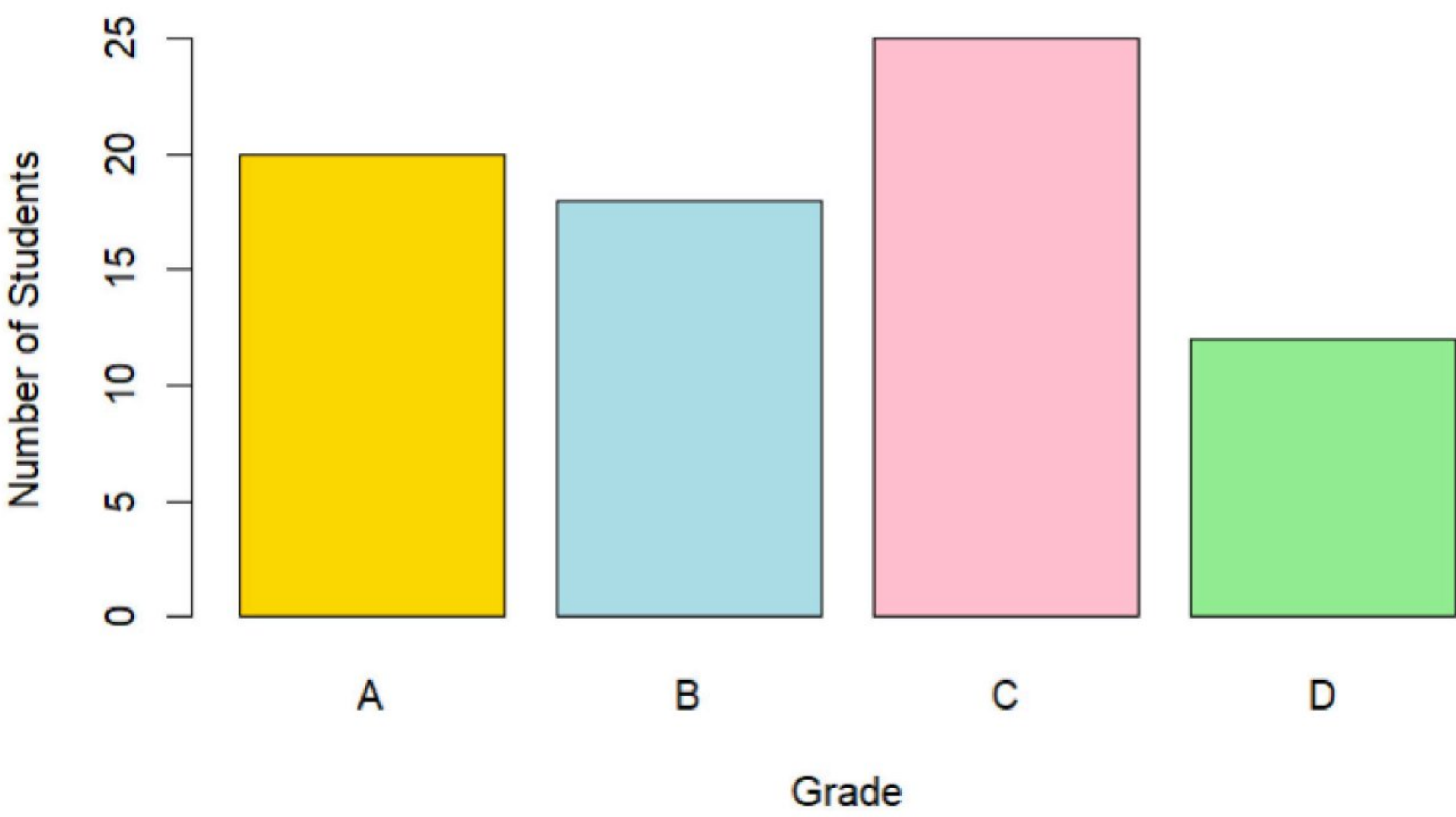
#PIECHART

Grade Distribution



#BARCHART

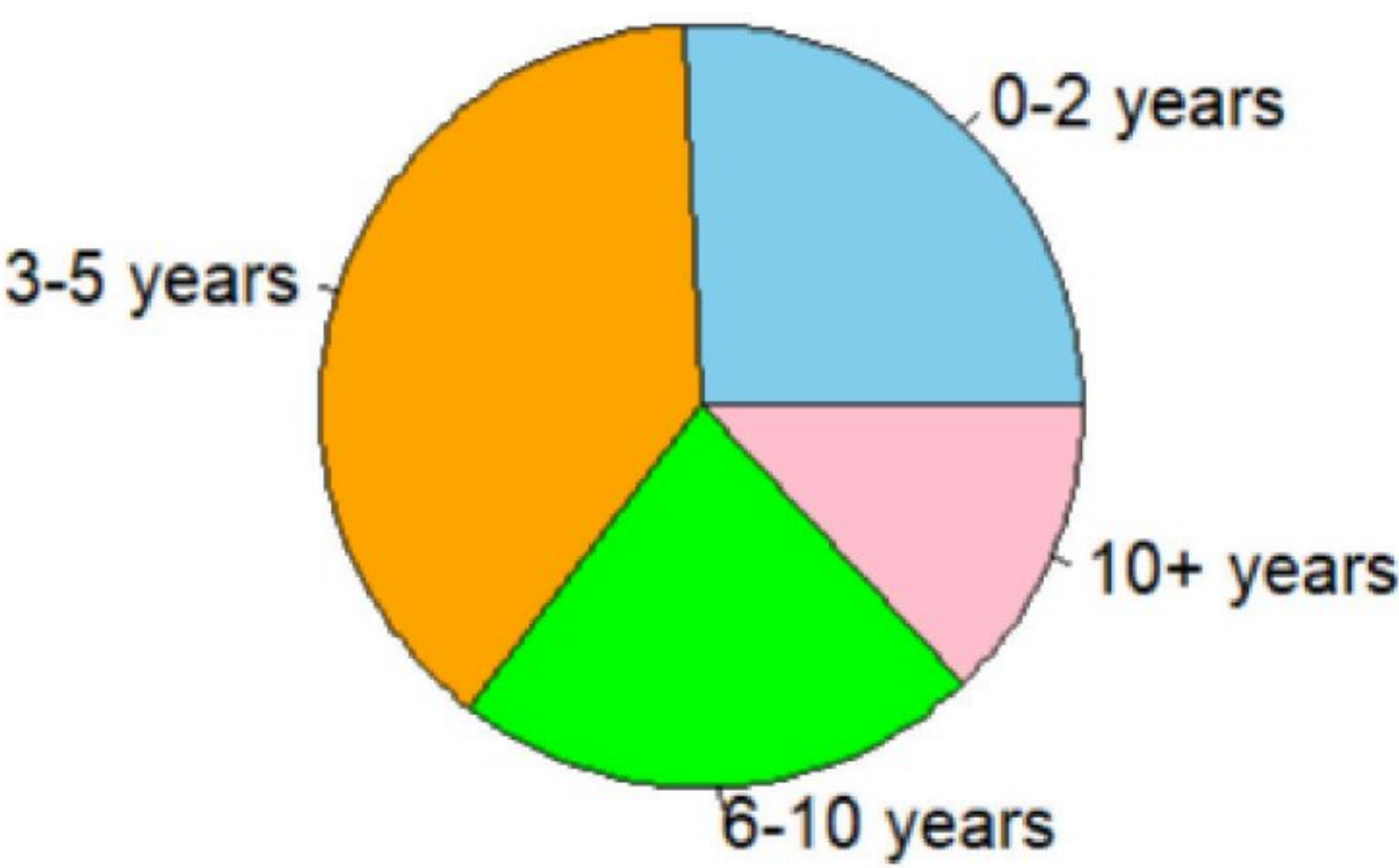
Number of Students per Grade



Question 4: Company Employee Experience Levels

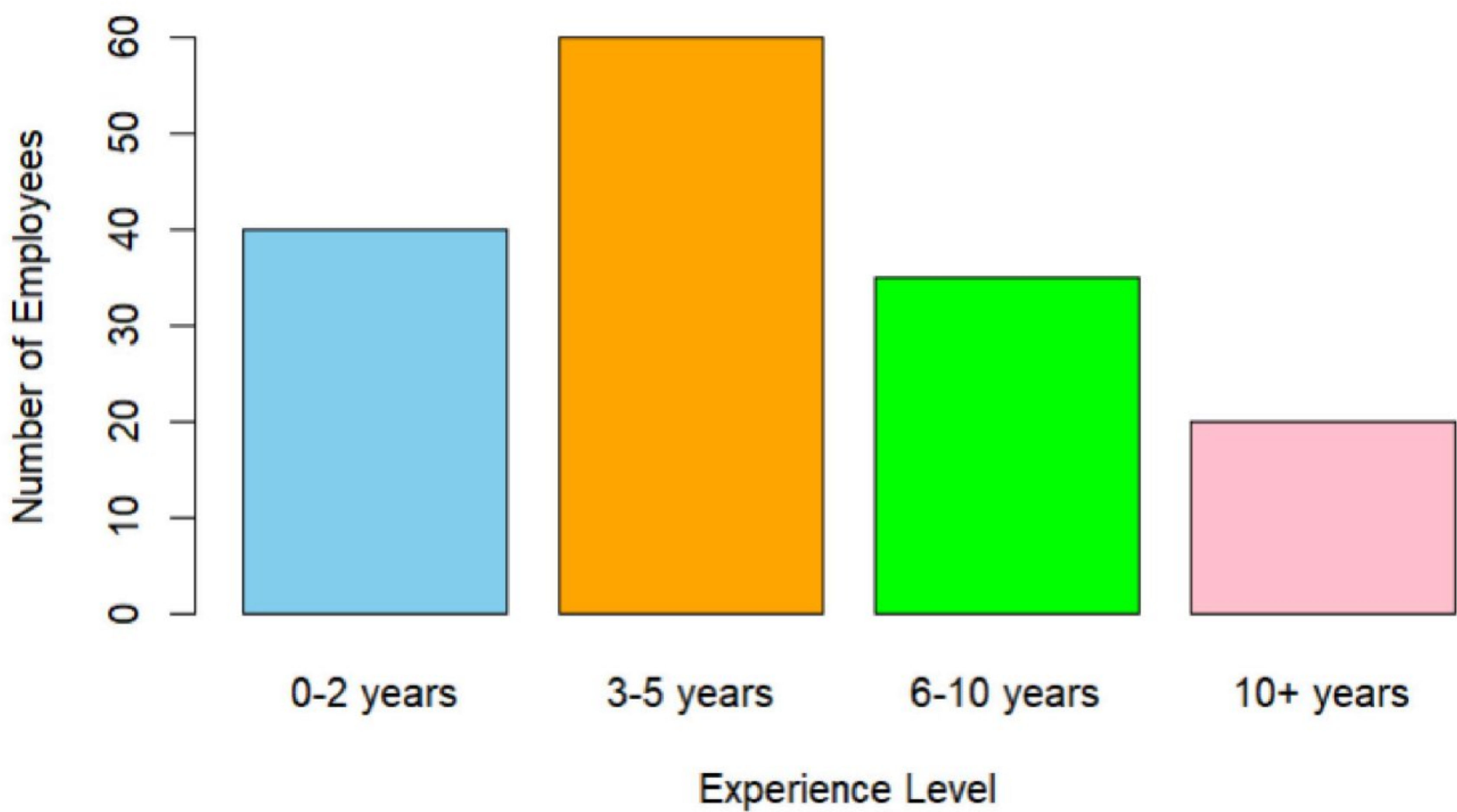
#PIECHART

Employee Experience Distribution



#BARCHART

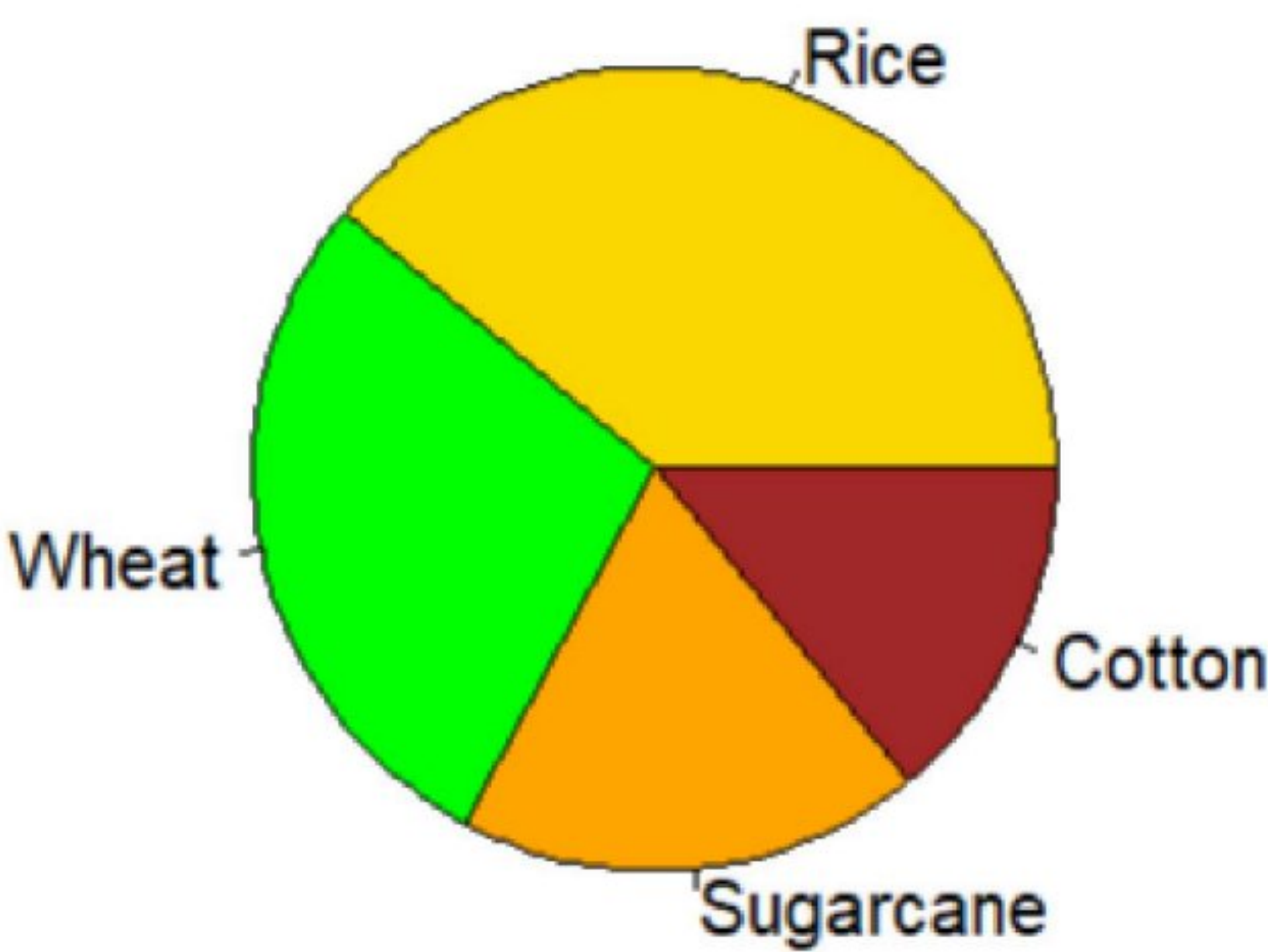
Number of Employees by Experience



Question 5: Agricultural Crop Production

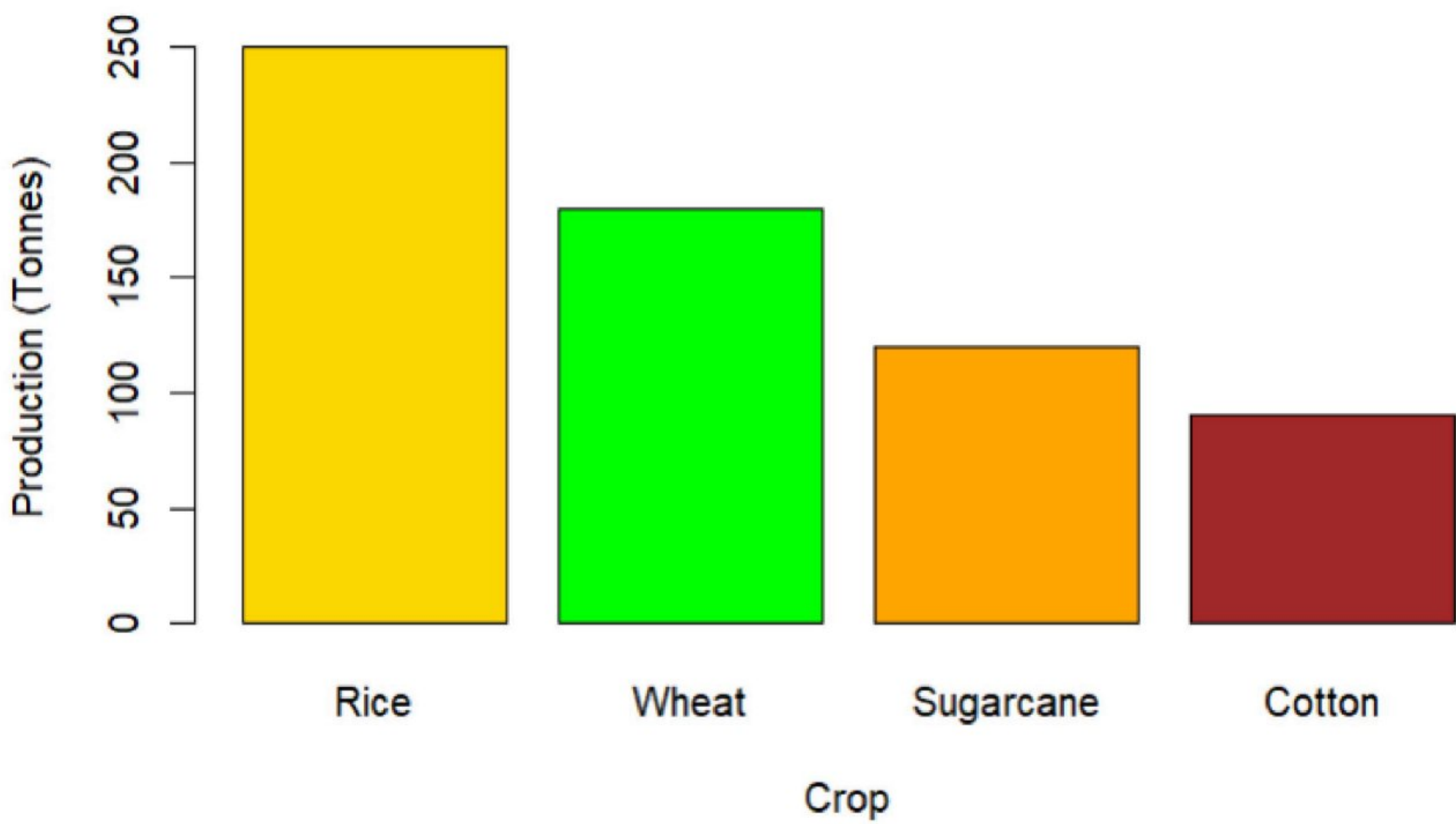
#PIECHART

Crop Production Distribution



#BARCHART

Crop Production in Tonnes



////////////////////////////////EXP-9////////////////////////////////

```
> # Question 1: School Student Marks Analysis
>
> students <- data.frame(
+   Name = c("Arun","Priya","Kiran","Meena","Rahul","Sneha","Vijay","Lakshmi","Suresh","Anu"),
+   Maths = c(78,85,90,88,76,92,81,79,95,84),
+   Science = c(80,89,85,90,72,94,78,83,91,86),
+   English = c(82,87,88,91,75,93,80,85,94,89)
+ )
>
> mean_marks <- sapply(students[,2:4], mean)
> median_marks <- sapply(students[,2:4], median)
> sd_marks <- sapply(students[,2:4], sd)
>
> students$Total <- rowSums(students[,2:4])
> top_student <- students[which.max(students$Total), ]
>
> avg_math <- mean(students$Maths)
> above_avg_math <- students[students$Maths > avg_math, ]
>
> mean_marks
  Maths Science English
  84.8    84.8    86.4
> median_marks
  Maths Science English
  84.5    85.5    87.5
> sd_marks
  Maths Science English
6.373556 6.713171 6.003703
> top_student
  Name Maths Science English Total
9 Suresh    95     91     94    280
> above_avg_math
  Name Maths Science English Total
2 Priya    85     89     87    261
3 Kiran    90     85     88    263
4 Meena    88     90     91    269
6 Sneha    92     94     93    279
9 Suresh    95     91     94    280
>
```

> # Question 2: Retail Sales Data Analysis

```
>
> sales <- data.frame(
+   Product = c("Mobile","Laptop","Mouse","Bag","Tablet","Camera","Speaker"),
+   Day1 = c(4,3,8,10,5,2,6),
+   Day2 = c(5,2,7,9,6,3,5),
+   Day3 = c(6,4,6,8,7,2,7),
+   Day4 = c(5,3,7,11,6,4,6),
+   Day5 = c(7,5,5,10,8,3,8)
+ )
>
> sales_stats <- data.frame(
+   Product = sales$Product,
+   Mean = apply(sales[,2:6], 1, mean),
+   Max = apply(sales[,2:6], 1, max),
+   Min = apply(sales[,2:6], 1, min),
+   Variance = apply(sales[,2:6], 1, var)
+ )
>
> max_avg_product <- sales_stats[which.max(sales_stats$Mean), ]
>
> sales_stats
  Product Mean Max Min Variance
1 Mobile  5.4   7   4   1.3
2 Laptop  3.4   5   2   1.3
3 Mouse  6.6   8   5   1.3
4 Bag    9.6  11   8   1.3
5 Tablet 6.4   8   5   1.3
6 Camera 2.8   4   2   0.7
7 Speaker 6.4   8   5   1.3
> max_avg_product
  Product Mean Max Min Variance
4 Bag    9.6  11   8   1.3
>
```



```

>
> # Question 3: Hospital Patient Age Analysis
>
> patients <- data.frame(
+   PatientID = 301:310,
+   Name = c("Ravi", "Meena", "Arjun", "Sneha", "Kiran", "Priya", "Vijay", "Lakshmi", "Suresh", "Anu"),
+   Age = c(40, 55, 60, 35, 48, 52, 45, 38, 65, 50)
+ )
>
> mean_age <- mean(patients$Age)
> median_age <- median(patients$Age)
> range_age <- range(patients$Age)
> sd_age <- sd(patients$Age)
>
> older_patients <- patients[patients$Age > 50, ]
>
> mean_age
[1] 48.8
> median_age
[1] 49
> range_age
[1] 35 65
> sd_age
[1] 9.647107
> older_patients
  PatientID   Name Age
2         302  Meena  55
3         303  Arjun  60
6         306  Priya  52
9         309 Suresh  65
~

> # Question 4: Employee Salary Analysis
>
> employees <- data.frame(
+   EmpID = 201:210,
+   Name = c("Arun", "Priya", "Kiran", "Meena", "Rahul", "Sneha", "Vijay", "Lakshmi", "Suresh", "Anu"),
+   Dept = c("IT", "HR", "Finance", "IT", "HR", "Finance", "IT", "HR", "Finance", "IT"),
+   Salary = c(35000, 38000, 45000, 36000, 39000, 47000, 34000, 40000, 50000, 37000)
+ )
>
> dept_stats <- aggregate(Salary ~ Dept, data=employees,
+   function(x) c(Mean=mean(x), Median=median(x)))
> dept_stats <- do.call(data.frame, dept_stats)
>
> overall_avg <- mean(employees$Salary)
> high_earners <- employees[employees$Salary > overall_avg, ]
>
> dept_stats
  Dept Salary.Mean Salary.Median
1 Finance  47333.33         47000
2    HR    39000.00         39000
3    IT    35500.00         35500
> high_earners
  EmpID   Name   Dept Salary
3   203  Kiran Finance  45000
6   206  Sneha Finance  47000
9   209 Suresh Finance  50000
~

```



```

<
> # Question 5: Online Store Customer Ratings
>
> ratings <- data.frame(
+   ProductID = 1:10,
+   Product = c("Phone", "Laptop", "Mouse", "Bag", "Camera", "Speaker", "Tablet", "Watch", "Charger", "Keyboard"),
+   Rating = c(5, 4, 3, 5, 4, 5, 4, 3, 5, 4)
+ )
>
> mean_rating <- mean(ratings$Rating)
> median_rating <- median(ratings$Rating)
> var_rating <- var(ratings$Rating)
>
> getmode <- function(v){
+   uniqv <- unique(v)
+   uniqv[which.max(tabulate(match(v, uniqv)))]
+ }
>
> mode_rating <- getmode(ratings$Rating)
>
> top_rated <- ratings[ratings$Rating > 4, ]
>
> mean_rating
[1] 4.2
> median_rating
[1] 4
> mode_rating
[1] 5
> var_rating
[1] 0.6222222
> top_rated
  ProductID Product Rating
1         1   Phone      5
4         4     Bag      5
6         6 Speaker      5
9         9 Charger      5
> |

```